

**Integration of Ems and Eds Systems in High-Speed Rail Transportation in China: A
Selective Literature Review**

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High-speed rail has become one of the most significant advancements in modern transportation, dramatically increasing mobility, environmental performance, and regional economic growth (Tang & Zhao, 2025). China is a global leader in this high-speed rail technology, particularly in magnetic levitation (Maglev) trains, which can reach extremely high speeds while effectively reducing friction and energy loss. Maglev trains rely on advanced suspension technologies to lift and stabilize the train above the track without physical contact. This paper is a selective literature review of existing research on the integration of two primary Maglev suspension systems: the Electro-Magnetic Suspension (EMS) and the Electro-Dynamic suspension (EDS).

Despite significant advances in theoretical modeling and component development, current research on maglev is often fragmented and incomplete. Existing academic literature often looks at EMS and EDS separately, focusing on system-specific issues like vibration suppression, mechanical coupling-a mechanism used to connect to shafts to transmit power-, and adjusting the control of a dynamic system (Wang et al., 2025). This paper argues that the absence of integrated, real-world operational data is the central technological barrier preventing the deployment of next-generation Maglev systems. Therefore, this proposal seeks to directly address the urgent need for operationally ready hybrid models. This is achieved by centering the inquiry around the critical research directly investigating in what ways has the integration of Electro-Magnetic Suspension (EMS) and Electro-Dynamic Suspension (EDS) systems impacted high-speed rail transportation in China?

Successfully answering this question will drive significant, measurable change across several domains. Technologically, a validated hybrid model will dramatically reduce the incidence of systems Failures by eliminating the current weakness inherent in transitional speed

ranges. Operationally, it will lower the total energy consumed per kilometer by optimizing the use of EMS and EDS, reducing long-term operational costs for railway authorities. Strategically, this research will provide policymakers with the necessary data to justify large scale sustainable public infrastructure investments (Tang & Zhao, 2025).

To build the case for this necessary research, this paper first reviews knowledge on component design, control systems, and economic viability. Following the review, the paper presents a methodology detailing the full-scale testing and data analysis protocols required to fill the operational knowledge gap. Finally, the addendum outlines the subsequent research steps necessary for final commercial readiness.

Literature Review

Answering the central research question requires a synthesis of learning across three distinct areas that define the difficulty of system integration. This review begins by examining the core engineering requirements necessary to overcome the physical challenge of combining the two suspension methods. Moving from the foundation principles, it then moves to the control system innovations needed to guarantee the integrated systems reliability and resilience. Finally, the analysis concludes by assessing the economic viability and sustainability required to justify large-scale deployment. By organizing the current research in this manner, this review builds the necessary case for the system level research needed to fill the gap between theoretical models and operational readiness.

Technological Foundations of EMS and EDS

Understanding the core mechanical and electromagnetic principles is essential to understand the complex technical requirements of hybrid integration. The EMS system is defined

by its ability to achieve precise levitation control through continuously adjusting the attractive force of electromagnets acting against a ferromagnetic—materials strongly attracted to magnets—rail (Chen et al., 2025). This active control loop, while offering high stability at low speeds, makes the system open to minor disturbances and requires continuous energy input to maintain the levitation gap. Chen et al. (2025) explores the stability challenges posed by coupling EMS with flexible train frames, talking mainly about the need for highly sophisticated mechanical-electromagnetic modeling to prevent resonance and instability. In contrast, the EDS system uses powerful superconducting magnets mounted on the train to induce currents in coiled along the guideway, generating a repulsive force that passively stabilizes the train once a certain lift-off speed is achieved (Wang et al., 2025). While this provides inherent high-speed stability, it requires a significant production of lower temperatures and energy cooling, alongside needing wheels or an auxiliary system for starting and stopping. These inherent differences mean that designers must make strategic trade-offs when selecting a maglev system, leading to compromising that affects performance across different speed ranges.

This leads directly to the concept of hybrid integration, a strategy designed to actively combine the systems to offset the drawbacks of one while utilizing the strength of the other. Researchers are exploring structures where EMS handles low-speed levitation and fine-tuning control, while EDS provides high-speed lift and strong stability. Huang et al. (2025) demonstrates that a hybrid superconducting-electromagnetic approach can significantly reduce vibration and enhance energy efficiency compared to using either system in isolation. This cooperation potentially creates a more resilient system capable of maintaining stability across all operational speeds. Furthermore, the hybrid model introduces complex challenges in control for a short time—the smooth transition between EMS-dominant and EDS-dominant operation

(Schiefer et al., 2025). Achieving this integration requires sophisticated solutions, including seamless switching mechanisms and continuous real-time data fusion, demanding innovation in smart electromechanical coupling and material science (Huang et al., 2025). However, despite promising laboratory data, the lack of full-scale track testing across China's diverse environmental conditions remains the main reason stopping the integration of the solutions.

Control Systems, Fault Tolerance, and the Application of AI

Moving beyond the mechanical challenge of dynamic switching, which impacts core systems stability, this review must talk about the impact of integration on operational safety and resilience. The difficulty of hybrid EMS-EDS systems elevates the importance of strategies, particularly in ensuring system reliability and fault tolerance—a critical aspect for safe public transportation. The control system manages not only the individual levitation gaps but also the dynamic interactions between the attractive (EMS) and repulsive (EDS) forces, especially during the high-speed transition phases.

The use of Artificial Intelligence (AI) in control systems has become a central research topic. Zhou et al. (2025) highlights the necessity of fault-tolerant control (FTC) to address concurrent sensor and actuator faults, which are more likely in a complex environment. Their research proposes a novel approach using radial basis function (RBF) neural networks to estimate and compensate for system uncertainty, allowing the levitation system to maintain stability even with component degradation. This reliance on AI-driven compensation is an essential requirement for making it more reliable for future Chinese maglev networks. Furthermore, energy management within the control loop is vital for improving overall system efficiency. Xia et al. (2025) introduced an energy flow-based switching control model for electromagnetic dampers, demonstrating how power can be dynamically routed to minimize

waste while ensuring continuous stability. This strategy moves beyond simple levitation control by integrating energy efficiency into the fundamental logic, which is crucial given the high energy demands of both EMS and EDS components.

Despite these advancements, the leap from highly controlled simulated or laboratory environments to operational rail networks presents significant hurdles. Current research often assumes a stable electromagnetic environment, yet in reality, high speed travel creates complex aerodynamic and electromagnetic disturbances (Wang et al., 2025). Scalability in control systems means not just handling more components but adapting to the real time variability of network conditions including switch points and track curvature. The control strategies developed by Wang et al. (2025) for cross-coupled energy-efficient EDS, while powerful in isolation, must be adapted to co-exist with the active control requirements of the EMS component. Consequently, the combination of these control models—combining passive stability at speed with active position at low speeds—remains the greatest challenge for large scale fault control implementation in China.

Economic and Energy Efficiency, and Sustainability

Having established the technological and safety implications of system integration, the final consideration is the long-term viability and strategic impact on China's economy. The pursuit of hybrid maglev systems is completely linked to China's national policy goals for sustainable infrastructure and economic growth. Energy consumption is the largest operational expense and environmental impact for HSR. The integrated EMS-EDS approach is investigated as a core solution to reduce levitation energy losses by strategically utilizing each system's most efficient operational range (Huang et al., 2025). Schiefer et al. (2025) have contributed to this optimization by developing highly accurate nonlinear magnetic circuit models for

superconducting systems. These models allow engineers to more precisely predict energy use, moving past simplified assumptions, and identifying true improvement opportunities in the EDS component. This improved modeling is essential for developing economically viable, high speed rail lines.

On a broader scale, the economic and environmental viability is assessed through the lens of policy and infrastructure planning. Tang and Zhao (2025) analyze the strategic role of maglev expansion in China, concluding that the development contributes significantly to national goals by enabling cleaner, high-efficiency transport that reduces reliance on fossil fuels in the mobility sector. The expansion of HSR systems, including advanced maglev, improves transportation links between cities, fuels regional economic growth and reduces travel times providing substantial economic benefits. However, the high initial investment and operational complexity of hybrid Maglev systems demand a look at if the cost will be worth it over a long period of time (comprehensive lifecycle cost analysis). Current literature is strong on technological potential but weaker on looking forward to looking at economics in the long-term and assessing its environmental impact (Tang & Zhao, 2025). The true sustainability of the integrated system depends not only on momentary energy efficiency but also in the long run, maintenance cost, and the disposal of superconducting magnets and different electronics. Bridging this gap requires collaboration between engineering science and economics, ensuring hybrid technology supports long-term ecological and sustainable goals.

In summary, the literature confirms that the theoretical integration of the EMS and EDS systems is essential for achieving next generation HSR's impacting technology, safety and economics. However. Current research remains fragmented. Research is isolated to component models, lacks validation under real world fault conditions, and is hindered by the absence of

empirical life cycle cost analysis (LCCA) data (Tang & Zhao, 2025). The fundamental gap is the lack of a comprehensive, validated system level operational model. This established deficit calls for a methodological shift: Future research must move away from proving integration is possible toward determining how the integrated system performs safely, reliably, and cost effectively on a full-scale test environment, directly providing the data required to fully answer the research question.

Methodology

Based on the critical gaps identified in the literature, the proposed research will employ a mixed-methods, interdisciplinary study to transition the EMS-EDS technology from theoretical modeling to operational readiness, thereby directly providing the missing system-level data necessary for full commercial deployment. This approach combines engineering simulation, controlled field testing, and policy analysis. The project is structured to first develop a full-scale hybrid EMS-EDS prototype based on controlled test track conditions. The population of tests will involve running the prototype under varying speed profiles, load capacities, and environmental conditions at a dedicated national test location in China. The estimated budget for prototype development and initial testing is substantial, requiring significant government and private sector funding to cover the high cost of superconducting materials and specialized engineers.

The core solution is the testing of the integrated system, requiring data collection that will be both quantitative and qualitative. The quantitative data will include real-time sensor readings captured at very fast intervals, focusing specifically on levitation gap stability, dynamic force distribution during switching events, and precise power consumption for both EMS and EDS components (Chen et al., 2025; Huang et al., 2025; Schiefer et al., 2025; Wang et al., 2025; Xia

et al., 2025). This empirical evidence is important for validating the theoretical energy models. In parallel, qualitative data will be gathered through maintenance logs, video records of component wear, and expert interviews providing context for system degradation and operational performance.

The analysis of this data will be conducted using advanced engineering modeling software and economic forecasting models to precisely predict long-term usability. Specifically, FEMA (Finite Element analysis) will be used to analyze structural integrity and vibration patterns, while regression analysis will connect power usage to environmental variables to project realistic operational costs (Chen et al., 2025; Schiefer et al., 2025; Wang et al., 2025; Xia et al., 2025). The justification for this comprehensive approach is that it is necessary to move beyond the limited theoretical data that currently constrains the field. The scope is limited to the testing of the hybrid control system and LCCA within the controlled test environment (Huang et al., 2025; Tang & Zhao, 2025). The project aims at filling the gap in real-world operational data and system resilience analysis that prevents commercial deployment.

Implications

The integration of EMS and EDS systems carries profound implications for the future of sustainable, high-performance rail transport. Technically, these hybrid systems promise to significantly improve train stability, energy efficiency, and fault tolerance by blending EMS's precise control with Ed's steady lift, leading to the creation of safer, smoother, and more reliable systems. From a policy perspective, hybrid maglev systems align directly with China's national goals for low-carbon, high-efficiency infrastructure (Tang & Zhao, 2025). They encourage regional economic development and improved connectivity, supporting broader urban planning

objectives. Understanding the dynamics of hybrid systems helps policymakers make informed decisions that ensure technology benefits both sustainability and public interest.

Given these widespread implications, moving forward, closing the research gap requires a lot of interdisciplinary collaboration. Future research must move beyond the lab and focus on full scale operational testing, long term energy analysis, and full lifetime assessment. This involves the integration of expertise from engineering, control systems, environmental science, and AI. Support from national programs like the National Natural Science Foundation of China (NSFC) is essential to put this from a plan to action (Huang et al., 2025). By addressing this challenge, China can cement its leading position in high-speed sustainable transportation, offering a valuable model for a global HSR development. This technological success and the lessons learned in large-scale integration could also significantly inform and shape the future development of high-speed rail projects in other countries, including the United States.

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Addendum

Timeline	Duration	Budget	Description
Phase 1: Getting Ready	January 2026- February 2026	\$750 (Paying staff, fixing the track, making the sensors work)	• Get the final signal to begin from the engineers and make sure the plan is set for testing on the national track. • Install an EMS/EDS control unit and hook it up to all the track sensors so it can start working. • Check if all the data loggers and energy measuring tools are ready and working correctly.
Phase 2: Running the Tests	March 2026- July 2026	\$1000 (Paying over bills, paying the engineering team, replacing broken parts)	• Run controlled and fast trips to see if the train switches smoothly between the two lifting systems (EMS and EDS). • Collect the Data: Record all the numbers on how stable the train is, how much it shakes, and how much power it uses at different speeds. • Put a bad sensor on the track to check if the FTC can keep the train safe.
Phase 3: Looking at the Results	August 2026 – November 2026	\$500 (Software fees for modeling, hiring some economics expert)	• Use computer programs (FEA for example) to check if the train structure is strong and to compare the shaking data to our expected stability models. • Use math and charts (like Regression Analysis and LCCA) to guess the long-term running costs and see how much energy was saved. • Make it better overall by making a more reliable working model based on all the data found.
Phase 4: Letting	December 2026	\$250 (Costs for printing	• Write a complete, detailed report that shows all the results, how we made the controls, and the LCCA report to show cost efficiency.

Others Know		and sharing, travel for talks)	• Share it with the big rail companies and funding groups by giving them advice on how to start building.
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